CS-Based Mathematics Curriculum
Course 2

Geometry and Introductory Trigonometry

A Programmer's Introduction

With exercises to solve by hand and then in code

Pilot Edition
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Preface: How to Use This Book

Every chapter runs the same way: read the core ideas, work the examples with the solutions covered, solve the exercises by hand, and only then write the code. The programming exercises reuse the hand exercises as test cases, so an unsolved exercise leaves you with nothing to test against.

Each chapter has four parts:

- **Core ideas**: the rules and definitions, with the computer science reason for each.
- **Worked examples**: cover the solution, attempt it, then compare.
- **Exercises**: numbered 1.1, 1.2 and so on; pencil, paper, no calculator, no code.
- **Programming exercises**: numbered C1.1, C1.2 and so on; Python 3, plain functions, `assert` statements as tests.

Rules that make the book work:

1. Do the exercises in one sitting before opening an editor. Check them against the answer key at the back of the book, and log every miss with the rule you broke.
2. Treat each programming exercise as a unit test of your understanding. Name the function exactly as given so later chapters can import it.
3. Keep all code in one folder, one file per chapter (`geo01.py`, `geo02.py`). Chapter 11 assembles them into the course project, a 2D scene you can transform and hit-test.
4. Use only the standard library. `math`, `fractions` and `random` are allowed; NumPy, SymPy and game-engine vector classes are not, because the point is to write what they do. Drawing may use matplotlib, an HTML canvas or SVG.
5. An exercise marked **(proof)** is answered in full sentences, not symbols alone.

The book covers the ten topics of Course 2 in the curriculum, in the same order, and ends with the course project and an exit review. It assumes Course 1: linear equations, systems, quadratics and functions are used without comment. Plan on about one chapter per week, with the last two weeks for the project.

Contents

Preface: How to Use This Book **ii**

1 Logic and Proof **1**
1.1 Core Ideas . . . 1
1.2 Worked Examples . . . 2
1.3 Exercises . . . 2
1.4 Programming Exercises . . . 3

2 Lines and Angles **4**
2.1 Core Ideas . . . 4
2.2 Worked Examples . . . 5
2.3 Exercises . . . 5
2.4 Programming Exercises . . . 6

3 Triangles **7**
3.1 Core Ideas . . . 7
3.2 Worked Examples . . . 8
3.3 Exercises . . . 8
3.4 Programming Exercises . . . 9

4 Coordinate Geometry **10**
4.1 Core Ideas . . . 10
4.2 Worked Examples . . . 11
4.3 Exercises . . . 11
4.4 Programming Exercises . . . 12

5 Transformations **13**
5.1 Core Ideas . . . 13
5.2 Worked Examples . . . 14
5.3 Exercises . . . 14
5.4 Programming Exercises . . . 15

6 Polygons and Area **16**
6.1 Core Ideas . . . 16
6.2 Worked Examples . . . 17
6.3 Exercises . . . 17
6.4 Programming Exercises . . . 17

7 Circles **19**
7.1 Core Ideas . . . 19
7.2 Worked Examples . . . 20
7.3 Exercises . . . 20
7.4 Programming Exercises . . . 21

8 Right-Triangle Trigonometry **22**
8.1 Core Ideas . . . 22
8.2 Worked Examples . . . 23
8.3 Exercises . . . 23
8.4 Programming Exercises . . . 24

9 The Unit Circle and Radians **25**
9.1 Core Ideas . . . 25
9.2 Worked Examples . . . 26
9.3 Exercises . . . 26
9.4 Programming Exercises . . . 27

10 Vectors in Two Dimensions **28**
10.1 Core Ideas . . . 28
10.2 Worked Examples . . . 29
10.3 Exercises . . . 29
10.4 Programming Exercises . . . 30

11 Course Project **31**
11.1 The Six Stages . . . 31
11.2 Acceptance Checklist . . . 32
11.3 Stretch Goals . . . 32

Exit Review **33**

Answers to Exercises **34**

Chapter 1

Logic and Proof

Geometry is the first place you meet formal proof: a chain of statements, each justified by a definition, a postulate or an earlier result. The same chain is what a correct program is, and the vocabulary here, hypothesis, conclusion, contrapositive, counterexample, is the vocabulary of preconditions, postconditions and failing tests.

1.1 Core Ideas

Conditionals. "If P, then Q" has hypothesis P and conclusion Q, written $P \Rightarrow Q$. It claims nothing about cases where P is false. A function contract is a conditional: if the precondition holds, then the postcondition holds.

Converse, inverse, contrapositive. From $P \Rightarrow Q$: the converse is $Q \Rightarrow P$, the inverse is $\neg P \Rightarrow \neg Q$, and the contrapositive is $\neg Q \Rightarrow \neg P$. A statement and its contrapositive are logically equivalent (both true or both false). The converse and inverse are equivalent to each other, but not to the original. Confusing a statement with its converse is the single most common error in reasoning.

Biconditionals and definitions. "P if and only if Q" means both $P \Rightarrow Q$ and $Q \Rightarrow P$. Every definition is a biconditional: a right angle is an angle that measures 90°, and any angle that measures 90° is a right angle.

Deductive reasoning. Postulates (axioms) are accepted without proof. Theorems are proved from postulates, definitions and earlier theorems. A proof applies rules to given facts until the conclusion appears; it never appeals to a picture looking right.

Proof formats. A two-column proof lists statements on the left and the reason for each on the right. A paragraph proof is the same chain in prose. Common reasons: definition of midpoint, segment addition postulate ($AB + BC = AC$ when B is between A and C), angle addition postulate, substitution, and the reflexive, symmetric and transitive properties of equality.

Counterexamples. A universal claim is disproved by one case where the hypothesis holds and the conclusion fails. One failing test disproves "this function is correct" the same way; no number of passing tests proves it.

1.2 Worked Examples

Example 1.1 Write the converse, inverse and contrapositive of "If a shape is a square, then it has four right angles," and say which are true.

Solution. Converse: if a shape has four right angles, then it is a square (false; a non-square rectangle is a counterexample). Inverse: if a shape is not a square, then it does not have four right angles (false, same counterexample). Contrapositive: if a shape does not have four right angles, then it is not a square (true, like the original).

Example 1.2 Given that M is the midpoint of $\overline{AB}$ and $AB = 10$, prove $AM = 5$.

Solution.

1. M is the midpoint of $\overline{AB}$. (Given)
2. $AM = MB$. (Definition of midpoint)
3. $AM + MB = AB$. (Segment addition postulate)
4. $AM + AM = 10$. (Substitution, using steps 2 and 3 and $AB = 10$)
5. $2\,AM = 10$, so $AM = 5$. (Combine like terms, then divide both sides by 2)

Example 1.3 Disprove: "If $x^2 = 9$, then $x = 3$."

Solution. $x = -3$ satisfies the hypothesis ($(-3)^2 = 9$) but not the conclusion. One counterexample is enough.

1.3 Exercises

1.1 Identify the hypothesis and conclusion: "If two angles are complementary, then their measures add to 90°."

1.2 Write the converse, inverse and contrapositive of "If two angles are vertical angles, then they are congruent." Which of the four statements are true?

1.3 Give a counterexample to "If a number is divisible by 2, then it is divisible by 4."

1.4 Points A, B, C, D lie on a line in that order, and $AB = CD$. Prove $AC = BD$.

1.5 $\angle 1$ and $\angle 2$ are supplementary, and $\angle 2$ and $\angle 3$ are supplementary. Prove $\angle 1 \cong \angle 3$.

1.6 Rewrite the definition of a right angle as a biconditional, then as two separate conditionals.

1.7 Is "If $x > 5$, then $x > 3$" true? Is its converse true? Give a counterexample if not.

1.8 State the contrapositive of "If a triangle has two congruent angles, then it is isosceles," and explain why proving the contrapositive would prove the original.

1.9 A function has precondition $n \geq 0$ and postcondition "returns $n!$." Write the contract as a conditional and its contra-

positive. What does the contrapositive say about a call that returned the wrong value?

1.10 Given "All squares are rectangles" and "$ABCD$ is a square," what can you conclude? Given instead "$ABCD$ is a rectangle," can you conclude it is a square? Name the error in reasoning that would.

1.4 Programming Exercises

C1.1 Write `implies(p, q)` returning the truth value of $p \Rightarrow q$ (false only when p is true and q is false). Print a truth table over all four combinations for the statement, its converse, inverse and contrapositive, and confirm which columns match.

C1.2 Write `find_counterexample(hypothesis, conclusion, candidates)` that returns the first candidate where the hypothesis holds and the conclusion fails, or `None`. Use it on Exercise 1.3 with lambdas over the integers 1 to 100.

C1.3 Write `factorial(n)` that asserts its precondition and, in a wrapper, asserts the postcondition $n! = n \cdot (n-1)!$ for $n \geq 1$. Show that a call violating the precondition fails at the assertion rather than returning a wrong value.

C1.4 Represent a two-column proof as a list of `(statement, reason, depends_on)` tuples, where `depends_on` lists earlier line numbers. Write `check_proof(lines, allowed_reasons)` that rejects any reason not in the allowed set and any dependency on a later line. Encode the proof in Example 1.2.

Chapter 2

Lines and Angles

Angles are the first quantities geometry measures, and the relationships between them, around a point, along a line, and across a pair of parallel lines, are the first theorems you prove. A layout engine, a ray cast through parallel lanes, and an angle normalized to one turn all run on exactly these facts.

2.1 Core Ideas

Vocabulary. A segment $\overline{AB}$ has two endpoints; a ray $\overrightarrow{AB}$ has one; a line $\overleftrightarrow{AB}$ has none. An angle is formed by two rays with a common endpoint, the vertex, and is measured in degrees: acute below 90°, right at 90°, obtuse between 90° and 180°, straight at 180°.

Angle pairs. Complementary angles sum to 90°; supplementary angles sum to 180°. Adjacent angles share a vertex and a side. A linear pair is two adjacent angles whose outer sides form a line; they are supplementary. Vertical angles are the opposite angles formed by two crossing lines; they are congruent.

Parallel lines and a transversal. When a transversal crosses two parallel lines (Figure 2.1), corresponding angles are congruent ($\angle 1 \cong \angle 5$), alternate interior angles are congruent ($\angle 3 \cong \angle 6$), alternate exterior angles are congruent ($\angle 1 \cong \angle 8$), and same-side interior angles are supplementary ($\angle 3 + \angle 5 = 180^\circ$). The converses are how you prove two lines are parallel.

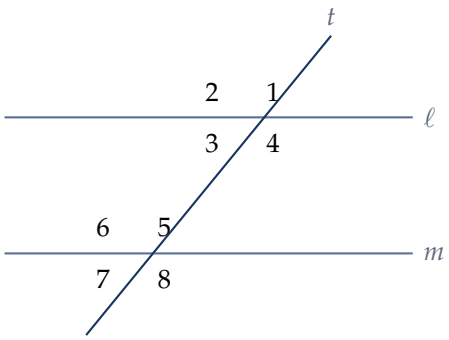


Figure 2.1: Two parallel lines ℓ and m cut by a transversal t. Angles 1, 2, 7, 8 are exterior; 3, 4, 5, 6 are interior.

Perpendicular lines. Two lines that meet at a right angle. In the coordinate plane their slopes multiply to -1, as in Course 1.

Angles of a triangle. The three interior angles of a triangle sum to 180°. The proof draws a line through one vertex parallel to the opposite side and uses alternate interior angles. An exterior angle of a triangle equals the sum of the two remote interior angles.

Angles in code. Angles are stored as numbers and must be normalized, usually to $[0^\circ, 360^\circ)$, because 735° and 15° are the same direction. The same facts hold for radians, which Chapter 9 introduces.

2.2 Worked Examples

Example 2.1 Two vertical angles measure $3x + 10$ and $5x - 30$. Find x and the angles.

Solution. Vertical angles are congruent, so $3x + 10 = 5x - 30$, which gives $2x = 40$ and $x = 20$. Each angle measures $3(20) + 10 = 70^\circ$.

Example 2.2 In Figure 2.1, $\angle 3 = 115^\circ$. Find all eight angles.

Solution. $\angle 2$ is vertical to $\angle 3$, so $\angle 2 = 115^\circ$. $\angle 1$ and $\angle 4$ are supplementary to $\angle 3$ (linear pairs), so each is 65°. By corresponding angles, $\angle 5 = \angle 1 = 65^\circ$, $\angle 6 = \angle 2 = 115^\circ$, $\angle 7 = \angle 3 = 115^\circ$ and $\angle 8 = \angle 4 = 65^\circ$.

Example 2.3 The angles of a triangle measure $2x$, $3x$ and $4x$. Find them.

Solution. $2x + 3x + 4x = 180$, so $9x = 180$ and $x = 20$. The angles are 40°, 60° and 80°.

2.3 Exercises

2.1 Find the complement and the supplement of a 37° angle.

2.2 Two vertical angles measure $4x - 15$ and $2x + 25$. Find x and the angle measure.

2.3 A linear pair measures $3x + 12$ and $2x - 7$. Find x and both angles.

2.4 Two parallel lines are cut by a transversal. A pair of corresponding angles measure $2x + 30$ and $4x - 10$. Find x and the angle.

2.5 Same-side interior angles measure $x + 40$ and $3x$. Find both angles.

2.6 A triangle has angles 50°, $2x$ and $3x - 20$. Find the two unknown angles.

2.7 An exterior angle of a triangle measures 120° and one remote interior angle measures 45°. Find the other remote interior angle.

2.8 **(proof)** Prove that vertical angles are congruent. Use that each of the two angles forms a linear pair with the same third angle.

2.9 Two lines are cut by a transversal. In the first case the alternate interior angles are 62° and 118°; in the second they are 118° and 118°. In which case are the lines parallel, and why?

2.10 Normalize 735° and -90° to the range $[0^\circ, 360^\circ)$.

2.11 **(proof)** Prove that the interior angles of a triangle sum to 180° by drawing a line through one vertex parallel to the opposite side.

2.4 Programming Exercises

C2.1 Write `normalize_angle(deg)` that returns the equivalent angle in $[0, 360)$ using only addition and subtraction in a loop, then a second version using `%`. Assert they agree for every integer from -1000 to 1000, and check Exercise 2.10.

C2.2 Write `transversal_angles(a1)` that takes the measure of $\angle 1$ in Figure 2.1 and returns a dictionary of all eight angles, assuming the lines are parallel. Check it against Example 2.2 by passing 65.

C2.3 Write `third_angle(a, b)` that returns the third angle of a triangle and raises `ValueError` when $a + b \geq 180$ or either input is not positive. Test with Exercise 2.6.

C2.4 Write `classify_pair(a, b)` returning `"complementary"`, `"supplementary"` or `"neither"`, using `math.isclose` so that $0.1 + 89.9$ counts as complementary.

C2.5 Draw two parallel lines and a transversal with matplotlib, SVG or a canvas, and label the eight angles with the values from C2.2. Change the transversal's slope and confirm the labels still satisfy every relationship in the Core Ideas.

Chapter 3

Triangles

Every mesh a graphics card draws is made of triangles, every distance on a screen is a hypotenuse, and every collision test begins with a triangle inequality. This chapter gives you the facts that make triangles rigid, the tests that make two of them congruent or similar, and the theorem that turns right triangles into distance.

3.1 Core Ideas

Classification. By sides: scalene (no equal sides), isosceles (at least two), equilateral (all three). By angles: acute (all less than 90°), right (one of 90°), obtuse (one greater than 90°). An equilateral triangle is equiangular, with three 60° angles.

Triangle inequality. Three lengths form a triangle only if each is less than the sum of the other two: $a + b > c$, $a + c > b$, $b + c > a$. Equality gives a degenerate "triangle" with zero area. Validate with this before constructing anything.

Congruence. Two triangles are congruent when corresponding sides and angles match. Four shortcuts prove it: SSS (three sides), SAS (two sides and the included angle), ASA (two angles and the included side) and AAS (two angles and a non-included side), plus HL for right triangles (hypotenuse and a leg). SSA and AAA do not prove congruence. Once triangles are congruent, every corresponding part is congruent (CPCTC).

Isosceles triangles. The angles opposite the two equal sides are equal, and the converse holds. The altitude from the vertex angle bisects the base and the angle.

Similarity. Triangles are similar when corresponding angles are equal and corresponding sides are proportional by one scale factor k. AA is enough to prove it. Perimeters scale by k and areas by k^2. Scaling a sprite, a camera projection and a shadow are all similar triangles.

The Pythagorean theorem. In a right triangle with legs a, b and hypotenuse c, the squares of the legs sum to the square of the hypotenuse. The converse is also true, which makes it a test for right angles.

Pythagorean theorem and special right triangles

$$a^2 + b^2 = c^2$$

A 45°-45°-90° triangle has sides in the ratio $1 : 1 : \sqrt{2}$. A 30°-60°-90° triangle has sides in the ratio $1 : \sqrt{3} : 2$, with the shortest side opposite the 30° angle.

A proof by rearrangement. Place four copies of the right triangle inside a square of side $a+b$ so that their hypotenuses form a tilted inner square of side c (Figure 3.1). The big square's area is $(a+b)^2 = a^2 + 2ab + b^2$; it is also c^2 plus four triangles of area $\tfrac{1}{2}ab$, which is $c^2 + 2ab$. Subtracting $2ab$ from both gives the theorem.

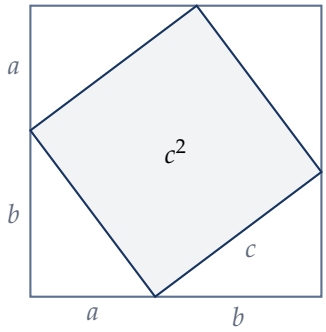


Figure 3.1: Four congruent right triangles inside a square of side $a+b$ leave a square of side c.

3.2 Worked Examples

Example 3.1 Can 4, 7 and 12 be the sides of a triangle?

Solution. $4+7=11$, which is not greater than 12, so no. The two short sides cannot reach each other.

Example 3.2 A right triangle has legs 9 and 12. Find the hypotenuse. Another has hypotenuse 13 and one leg 5. Find the other leg.

Solution. $c = \sqrt{81+144} = \sqrt{225} = 15$. For the second, $b = \sqrt{169-25} = \sqrt{144} = 12$.

Example 3.3 A 3-4-5 triangle is enlarged so that its shortest side is 9. Find the other sides and the ratio of the areas.

Solution. The scale factor is $9/3 = 3$, so the sides are 9, 12 and 15. The area ratio is $3^2 = 9$.

3.3 Exercises

3.1 Classify by sides and by angles: (a) sides $5, 5, 5$; (b) angles $90^\circ, 45^\circ, 45^\circ$; (c) sides $7, 8, 12$ with all angles acute.

3.2 Which can be the sides of a triangle: (a) $3, 4, 8$; (b) $5, 5, 9$; (c) $2, 2, 4$?

3.3 Two sides of a triangle are 7 and 10.

Write the range of possible lengths for the third side.

3.4 A right triangle has legs 8 and 15. Find the hypotenuse.

3.5 A right triangle has hypotenuse 25 and one leg 7. Find the other leg.

3.6 Is a triangle with sides $9, 40, 41$ a right triangle? Is one with sides $7, 9, 12$? If not, is it acute or obtuse?

3.7 A $45°$-$45°$-$90°$ triangle has legs of 6; find the hypotenuse. A $30°$-$60°$-$90°$ triangle has shortest side 5; find the other two sides.

3.8 An isosceles triangle has a vertex angle of $40°$. Find the base angles.

3.9 $\triangle ABC \sim \triangle DEF$ with $AB = 6$, $DE = 9$ and $BC = 8$. Find EF and the ratio of the perimeters.

3.10 Which congruence shortcut applies, if any: (a) three pairs of congruent sides; (b) two angles and the included side; (c) two sides and a non-included angle?

3.11 A 13-foot ladder reaches a window 12 feet above the ground. How far is the base of the ladder from the wall?

3.12 **(proof)** Prove the Pythagorean theorem using Figure 3.1, justifying why the inner figure is a square.

3.13 A 1920 by 1080 pixel screen has what diagonal length in pixels, to one decimal?

3.4 Programming Exercises

C3.1 Write `is_triangle(a, b, c)` using the triangle inequality, and `triangle_type(a, b, c)` returning a pair such as `("isosceles", "obtuse")`, where the angle type comes from comparing c^2 with $a^2 + b^2$ for the longest side c. Test with Exercises 3.1, 3.2 and 3.6.

C3.2 Write `hypotenuse(a, b)` and `leg(c, a)`, raising `ValueError` when a leg would not be shorter than the hypotenuse. Compare `hypotenuse` with `math.hypot` on 1000 random inputs.

C3.3 Write `pythagorean_triples(limit)` that returns every (a, b, c) with $a < b < c \le$ `limit` and $a^2 + b^2 = c^2$, by brute force. List the triples with $c \le 50$ and check that $(9, 40, 41)$ appears.

C3.4 Write `similarity_scale(t1, t2)` that takes two side-length triples and returns the scale factor if the triangles are similar (sort each, then check the three ratios are equal within tolerance) or `None`. Test with Example 3.3.

C3.5 Two circles with centers (x_1, y_1), (x_2, y_2) and radii r_1, r_2 overlap when the distance between centers is at most $r_1 + r_2$. Write `circles_overlap(...)` that decides this without calling `sqrt`, by comparing squared quantities, and explain in a comment why that is both valid and faster.

Chapter 4

Coordinate Geometry

Everything on a screen is a pair of numbers, so every geometric question becomes arithmetic on coordinates: how far apart, what is in between, is it inside. This chapter also handles the one convention that trips up every programmer, a y-axis that points down.

4.1 Core Ideas

Distance and midpoint. The distance between two points is the hypotenuse of the right triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$; the midpoint averages the coordinates.

Distance and midpoint

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \qquad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Slope again. Slope, parallel ($m_1 = m_2$) and perpendicular ($m_1 m_2 = -1$) carry over from Course 1 and are the tools for proving a quadrilateral is a parallelogram or a rectangle from its vertices.

Circles. A circle with center (h, k) and radius r is the set of points at distance r from the center: $(x - h)^2 + (y - k)^2 = r^2$. A point is inside, on, or outside according to whether its squared distance from the center is less than, equal to, or greater than r^2. No square root is needed for the test.

Completing the square. The expanded form $x^2 + y^2 + ax + by + c = 0$ is converted back to center-radius form by completing the square in x and in y, exactly as in Course 1, Chapter 9.

Screen coordinates. Math coordinates put the origin at the center with y increasing upward. Screens put the origin at the top-left with y increasing downward. For a canvas of width w and height h, a math point (x, y) lands at screen $(w/2 + x,\ h/2 - y)$. Forgetting the sign flip mirrors every rotation.

Coordinate proofs. Place the figure so the algebra is easy (a vertex at the origin, a side on an axis, lengths $2a$ and $2b$ so midpoints are integers), then prove the claim with the distance, midpoint and slope formulas.

Hit testing. A point is inside an axis-aligned rectangle when both coordinates lie within its bounds; inside a circle when the squared distance test passes. These two checks are the broad phase of every collision system.

4.2 Worked Examples

Example 4.1 Find the distance and midpoint between $(-3, 4)$ and $(5, -2)$.

Solution. $d = \sqrt{8^2 + (-6)^2} = \sqrt{100} = 10$. $M = \left(\frac{-3+5}{2}, \frac{4+(-2)}{2}\right) = (1, 1)$.

Example 4.2 Write the equation of the circle with center $(2, -1)$ and radius 5. Is $(5, 3)$ on it?

Solution. $(x-2)^2 + (y+1)^2 = 25$. For $(5, 3)$: $3^2 + 4^2 = 25$, so yes, it lies on the circle.

Example 4.3 Prove that the midpoint of the hypotenuse of a right triangle is equidistant from all three vertices.

Solution. Place the right angle at the origin with vertices $(0, 0)$, $(2a, 0)$ and $(0, 2b)$. The hypotenuse's midpoint is (a, b). Its distance to each vertex: to $(0, 0)$ is $\sqrt{a^2 + b^2}$; to $(2a, 0)$ is $\sqrt{a^2 + b^2}$; to $(0, 2b)$ is $\sqrt{a^2 + b^2}$. All three are equal.

4.3 Exercises

4.1 Find the distance between $(1, 2)$ and $(7, 10)$.

4.2 Find the midpoint of $(-4, 6)$ and $(2, -2)$.

4.3 A segment has midpoint $(3, 5)$ and one endpoint $(-1, 8)$. Find the other endpoint.

4.4 Find the perimeter of the triangle with vertices $(0, 0)$, $(6, 0)$ and $(6, 8)$.

4.5 Write the equation of the circle with center $(-3, 2)$ and radius 4.

4.6 Find the center and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

4.7 Is $(7, 1)$ inside, on, or outside the circle with center $(3, -2)$ and radius 5?

4.8 Show that $A(0, 0)$, $B(4, 0)$, $C(5, 3)$, $D(1, 3)$ form a parallelogram, once using slopes and once using the midpoints of the diagonals.

4.9 A canvas is 800 by 600 with the math origin at its center. Where does the math point $(-100, 50)$ land on screen? Which math point is at screen $(650, 450)$?

4.10 **(proof)** Prove that the diagonals of a rectangle are congruent, using vertices $(0, 0)$, $(a, 0)$, (a, b) and $(0, b)$.

4.11 A rectangle spans x from 10 to 60 and y from 20 to 40. Classify $(15, 25)$, $(60, 40)$ and $(70, 30)$ as inside, on the boundary, or outside.

4.4 Programming Exercises

C4.1 Write `distance(p, q)` and `midpoint(p, q)` on coordinate tuples. Compare `distance` with `math.dist` and check Exercises 4.1 to 4.3.

C4.2 Write `circle_from_general(a, b, c)` that converts $x^2 + y^2 + ax + by + c = 0$ into `(center, radius)` by completing the square in code, raising if the radius would be imaginary. Test with Exercise 4.6.

C4.3 Write `point_in_circle(p, center, r)` and `point_in_rect(p, x0, y0, x1, y1)` without square roots, each returning `"inside"`, `"on"` or `"outside"`. Test with Exercises 4.7 and 4.11.

C4.4 Write `to_screen(p, w, h)` and `to_math(s, w, h)`. Assert they round-trip for 1000 random points and reproduce Exercise 4.9.

C4.5 Write `is_parallelogram(a, b, c, d)` using the midpoints of the diagonals, then `is_rectangle(a, b, c, d)` that also checks adjacent sides are perpendicular by slope product. Test with Exercise 4.8 and with the rectangle in Exercise 4.10 for $a = 4$, $b = 3$.

Chapter 5

Transformations

A transformation is a function from points to points, and the four you meet here, translate, reflect, rotate and dilate, are the entire vocabulary of CSS transforms, canvas contexts and sprite movement. The two ideas that matter most are that composition is function composition and that its order matters.

5.1 Core Ideas

Translations. Slide every point by the same vector: $(x, y) \mapsto (x + a, y + b)$. Distances and angles are preserved.

Reflections. Across the x-axis: $(x, y) \mapsto (x, -y)$. Across the y-axis: $(x, y) \mapsto (-x, y)$. Across the line $y = x$: $(x, y) \mapsto (y, x)$. A reflection preserves distances but reverses orientation: a clockwise triangle becomes counterclockwise.

Rotations about the origin. Counterclockwise by $90°$: $(x, y) \mapsto (-y, x)$. By $180°$: $(x, y) \mapsto (-x, -y)$. By $270°$ counterclockwise, which is $90°$ clockwise: $(x, y) \mapsto (y, -x)$. Rotation by any angle θ is $(x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta)$; Chapters 8 and 9 supply the sine and cosine.

Isometries. Translations, reflections and rotations preserve every distance and angle, so the image is congruent to the original. A dilation by factor k about the origin, $(x, y) \mapsto (kx, ky)$, scales distances by $|k|$ and produces a similar figure, not a congruent one.

Composition and order. Applying f then g gives the function $g \circ f$, read right to left. Rotating then translating is not the same as translating then rotating. CSS reads `transform: translate(...) rotate(...)` the same way: the rightmost transform is applied to the element first.

Rotating about any point. To rotate about a center C, translate so C is at the origin, rotate, then translate back. Every "transform-origin" is this three-step composition.

Symmetry. A figure has line symmetry when a reflection maps it onto itself, and rotational symmetry of order n when a rotation by $360°/n$ does.

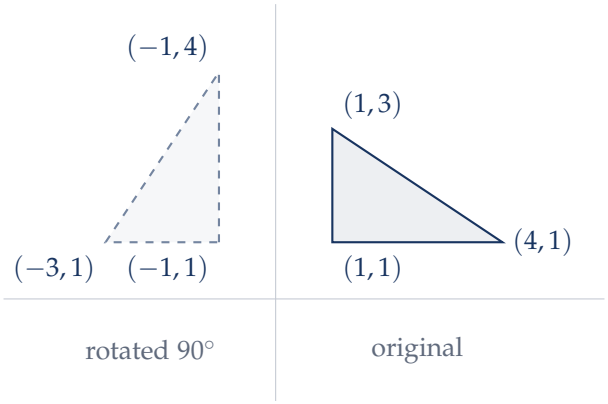


Figure 5.1: The triangle $(1,1)$, $(4,1)$, $(1,3)$ and its image under a 90° counterclockwise rotation about the origin, $(x, y) \mapsto (-y, x)$.

5.2 Worked Examples

Example 5.1 Translate the triangle $(1,1)$, $(4,1)$, $(1,3)$ by $(-2,3)$.

Solution. Add -2 to each x and 3 to each y: $(-1,4)$, $(2,4)$, $(-1,6)$.

Example 5.2 Rotate $(3,-2)$ about the origin by 90° counterclockwise and by 180°.

Solution. 90°: $(x, y) \mapsto (-y, x)$ gives $(2,3)$. 180°: $(-3,2)$.

Example 5.3 Reflect $(2,5)$ across the y-axis, then translate by $(1,-1)$. Then do the two steps in the other order.

Solution. Reflect first: $(-2,5)$, then translate: $(-1,4)$. Translate first: $(3,4)$, then reflect: $(-3,4)$. The results differ, so order matters.

5.3 Exercises

5.1 Translate $(-3,7)$ by $(5,-2)$.

5.2 Reflect $(4,-1)$ across the x-axis, the y-axis, and the line $y = x$.

5.3 Rotate $(-2,5)$ about the origin by 90° counterclockwise, by 180°, and by 90° clockwise.

5.4 Dilate $(6,-3)$ by $k = \frac{1}{3}$ about the origin. What scale factor maps $(2,3)$ to $(5,7.5)$?

5.5 Describe the transformation $(x, y) \mapsto (x-4, y+1)$ in words, then apply it to the square $(0,0)$, $(2,0)$, $(2,2)$, $(0,2)$.

5.6 Rotate $(4,2)$ by 90° counterclockwise about the point $(1,1)$.

5.7 Apply to $(3,1)$: reflect across the x-axis, then rotate 90° counterclockwise. Then reverse the order. Are the results the same?

5.8 Which of translation, rotation, reflection and dilation (with $k \neq \pm 1$) are isometries? Which reverse orientation?

5.9 What is the order of rotational symmetry of a regular hexagon, and the smallest angle of rotation that maps it to itself? How many lines of symmetry does a non-square rectangle have?

5.10 In CSS, where does a point at the element's origin end up under `translate(50px, 0) rotate(90deg)` versus `rotate(90deg) translate(50px, 0)`? Use screen coordinates, where a positive rotation turns clockwise.

5.11 **(proof)** Show that reflecting across $y = 0$ and then across $y = 3$ is a translation, and find it. State the general rule for two reflections across parallel lines a distance d apart.

5.4 Programming Exercises

C5.1 Write `translate(p, dx, dy)`, `reflect_x(p)`, `reflect_y(p)` and `reflect_diag(p)`, then `rotate90(p, times)` and finally `dilate(p, k)`, all on coordinate tuples. Test with Exercises 5.1 to 5.4.

C5.2 Write `compose(*fns)` that returns a function applying the given functions right to left. Reproduce Exercise 5.7 and show the two compositions disagree.

C5.3 Write `rotate90_about(p, center, times)` by composing translate, rotate and translate back. Test with Exercise 5.6.

C5.4 Represent each transformation as a 3×3 matrix on homogeneous coordinates $(x, y, 1)$, stored as nested lists. Write `apply(m, p)` and `multiply(a, b)`, and verify that multiplying matrices gives the same result as composing the functions from C5.2 on 100 random points.

C5.5 Draw a polygon and its images under a sequence of transforms (matplotlib, SVG or canvas), with the original in a different color, and confirm visually that rotating by 90° four times returns the original exactly.

Chapter 6

Polygons and Area

Polygons are the shapes a renderer actually fills, and area is the quantity that tells you how much it costs to fill them. The shoelace formula in this chapter computes any polygon's area from its vertices in one loop, and its sign tells you which way the vertices wind, which is how a graphics pipeline decides what is a front face.

6.1 Core Ideas

Angle sums. The interior angles of an n-sided polygon sum to $(n-2)\cdot 180^\circ$, because it splits into $n-2$ triangles from one vertex. Each interior angle of a regular n-gon is that sum divided by n. The exterior angles of any convex polygon sum to 360°, one full turn.

Quadrilaterals. A parallelogram has both pairs of opposite sides parallel; its opposite sides are equal and its diagonals bisect each other. A rectangle is a parallelogram with right angles; a rhombus is one with equal sides; a square is both. A trapezoid has exactly one pair of parallel sides. A kite has two pairs of adjacent equal sides and perpendicular diagonals.

Area formulas. Rectangle lw. Parallelogram bh, with h perpendicular to the base. Triangle $\frac{1}{2}bh$. Trapezoid $\frac{1}{2}(b_1+b_2)h$. Rhombus or kite $\frac{1}{2}d_1d_2$ from the diagonals. Regular polygon $\frac{1}{2}aP$, where a is the apothem (center to the middle of a side) and P the perimeter.

The shoelace formula. For a simple polygon with vertices $(x_1,y_1),\dots,(x_n,y_n)$ in order, with (x_{n+1},y_{n+1}) meaning the first vertex again, the signed area is given below. It is positive when the vertices run counterclockwise and negative when clockwise; take the absolute value for the geometric area.

Shoelace formula

$$A=\frac{1}{2}\sum_{i=1}^{n}\left(x_i\,y_{i+1}-x_{i+1}\,y_i\right)$$

Scaling. Scaling a figure by k multiplies lengths and perimeters by k and areas by k^2. Doubling a sprite's size quadruples the pixels drawn.

Bounding boxes. The axis-aligned bounding box of a polygon runs from its smallest to largest x and y. Two shapes whose boxes do not overlap cannot touch, which is why collision systems test boxes first and polygons only afterward.

6.2 Worked Examples

Example 6.1 Find the sum of the interior angles of a hexagon and each angle of a regular hexagon.

Solution. $(6-2)\cdot 180^\circ = 720^\circ$; each angle of a regular hexagon is $720^\circ/6 = 120^\circ$.

Example 6.2 Find the area of a trapezoid with bases 8 and 12 and height 5.

Solution. $\tfrac{1}{2}(8+12)(5) = 50$.

Example 6.3 Use the shoelace formula on the triangle $(0,0)$, $(4,0)$, $(0,3)$.

Solution. The terms are $0\cdot 0 - 4\cdot 0 = 0$, $4\cdot 3 - 0\cdot 0 = 12$, and $0\cdot 0 - 0\cdot 3 = 0$. Half the sum is 6, and the positive sign says the vertices run counterclockwise.

6.3 Exercises

6.1 Find the interior angle sum of an octagon, each interior angle of a regular octagon, and each exterior angle.

6.2 A regular polygon has interior angles of 150°. How many sides does it have?

6.3 Find the area of a parallelogram with base 9 and height 4, and of a triangle with base 10 and height 7.

6.4 Find the area of a trapezoid with bases 6 and 14 and height 5.

6.5 A rhombus has diagonals 10 and 24. Find its area and its side length.

6.6 Find the area of a regular hexagon with side 6, exactly and to one decimal.

6.7 Use the shoelace formula on $(1,1)$, $(5,1)$, $(6,4)$, $(2,5)$.

6.8 A rectangle is scaled by 3. By what factors do its perimeter and area change?

6.9 Find the area of the parallelogram $(0,0)$, $(5,0)$, $(7,3)$, $(2,3)$ using base times height, then confirm with the shoelace formula.

6.10 Find the bounding box of the triangle $(-2,4)$, $(5,-1)$, $(3,6)$ and its area, then the triangle's area by shoelace. What fraction of the box does the triangle fill?

6.11 **(proof)** Prove that the exterior angles of a convex n-gon sum to 360°, using the interior angle sum.

6.4 Programming Exercises

C6.1 Write `signed_area(points)` implementing the shoelace formula. Then write the function `is_counterclockwise(points)`, which reads the sign of that result. Test with Exercises 6.7, 6.9 and 6.10, and with each polygon's vertices reversed.

C6.2 Write `regular_polygon(n, side)` that returns the n vertices on a circle (use `math.cos` and `math.sin` with the angle $2\pi i/n$; Chapter 9 explains the radians). Compute its area with C6.1 and compare with $\tfrac{1}{2}aP$ for $n = 6$, side 6.

C6.3 Write `bounding_box(points)` and `boxes_overlap(a, b)`. Generate random triangles and confirm that whenever the boxes do not overlap, no vertex of one triangle lies inside the other.

C6.4 Write `interior_angle_sum(n)` and `sides_from_interior_angle(deg)`, raising when the angle does not correspond to a whole number of sides. Test with Exercises 6.1 and 6.2.

C6.5 Scale a polygon by k using `dilate` from Chapter 5 for $k = 1, 2, 3, 4, 5$, compute each area with C6.1, and print the ratio to the original to confirm it is k^2.

Chapter 7

Circles

Circles are how you draw anything round, lay items out radially, and test whether a ray hits a target. This chapter covers arcs and sectors, the angle theorems that make inscribed shapes predictable, and the algebra that finds where a line meets a circle.

7.1 Core Ideas

Parts of a circle. The radius r runs from the center to the circle; the diameter is $2r$. A chord joins two points on the circle; a secant is a line through two points; a tangent touches at exactly one point. Circumference is $2\pi r$ and area is πr^2.

Arcs and sectors. A central angle has its vertex at the center, and the arc it cuts off has the same measure in degrees. A central angle of θ degrees cuts off a fraction $\theta/360$ of the whole circle, which gives the arc length and the sector area.

Arc length and sector area (degrees)

$$s = \frac{\theta}{360} \cdot 2\pi r, \qquad A_{\text{sector}} = \frac{\theta}{360} \cdot \pi r^2$$

Inscribed angles. An inscribed angle has its vertex on the circle. It measures half the arc it intercepts, so it is half the central angle on the same arc. Any angle inscribed in a semicircle is a right angle, which is a quick way to construct one.

Tangents and chords. A tangent is perpendicular to the radius at the point of tangency, so a tangent segment, the radius, and the segment from the center to the external point form a right triangle. The two tangent segments from an external point are equal. A radius perpendicular to a chord bisects it.

Circle and line intersection. Substitute the line's equation into the circle's and solve the resulting quadratic. Two real solutions mean the line is a secant, one means it is tangent, none means it misses. The discriminant from Course 1 decides it before you solve.

Circles in code. Drawing APIs take a center, a radius and start and end angles, almost always in radians. A radial menu places n items at angles $360^\circ/n$ apart. A ray-circle test is the intersection above with the extra condition that the hit lies ahead of the ray's origin.

7.2 Worked Examples

Example 7.1 A circle has radius 10. Find the arc length and sector area for a central angle of 72°.

Solution. The fraction is $72/360 = 1/5$. Arc length: $\frac{1}{5}(20\pi) = 4\pi \approx 12.57$. Sector area: $\frac{1}{5}(100\pi) = 20\pi \approx 62.83$.

Example 7.2 An inscribed angle intercepts an arc of 100°. Find the angle.

Solution. Half the arc: 50°. A central angle on the same arc would be 100°.

Example 7.3 Where does the line $y = x + 1$ meet the circle $x^2 + y^2 = 25$?

Solution. Substitute: $x^2 + (x+1)^2 = 25$, so $2x^2 + 2x - 24 = 0$, or $x^2 + x - 12 = 0$, which factors as $(x+4)(x-3) = 0$. The points are $(3, 4)$ and $(-4, -3)$.

7.3 Exercises

7.1 Find the circumference and area of a circle with radius 7, exactly and to two decimals.

7.2 A circle has diameter 12. Find the arc length and sector area for a central angle of 60°.

7.3 An arc of length 5π lies on a circle of radius 10. Find its central angle.

7.4 An inscribed angle intercepts an arc of 140°. Find the inscribed angle and the central angle on the same arc.

7.5 A triangle is inscribed in a circle with one side as a diameter of length 13. One other side is 5. Find the third side and explain why you can.

7.6 A tangent segment from an external point P has length 8 and the radius is 6. How far is P from the center?

7.7 A chord of length 16 lies in a circle of radius 10. How far is the chord from the center?

7.8 Classify each line against the circle $x^2 + y^2 = 25$ as secant, tangent or miss, and give any intersection points: $y = 7$; $y = 5$; $y = 3$.

7.9 Find where the line $x + y = 7$ meets the circle $x^2 + y^2 = 25$.

7.10 Eight menu items are spaced equally on a circle of radius 120 pixels. What is the angle between neighbors, and what is the arc length between adjacent item centers?

7.11 A wheel of radius 0.35 m turns 500 times. How far does it travel, to one decimal?

7.12 **(proof)** Prove the inscribed angle theorem for the case where one side of the inscribed angle is a diameter, using an isosceles triangle and the exterior angle theorem.

7.4 Programming Exercises

C7.1 Write `arc_length(r, deg)`, `sector_area(r, deg)` and `angle_from_arc(r, s)`. Test with Exercises 7.2 and 7.3 and assert the first and third are inverses.

C7.2 Write `circle_line_intersections(center, r, p, q)` for the line through points `p` and `q`, returning a list of 0, 1 or 2 points. Parametrize the line as $p + t(q - p)$, substitute, and use the discriminant. Test with Exercises 7.8 and 7.9.

C7.3 Write `tangent_length(d, r)` for the tangent from a point at distance `d` from the center, and `chord_distance(r, chord)`, each raising on impossible inputs. Test with Exercises 7.6 and 7.7.

C7.4 Write `radial_positions(n, r, center)` returning the n points of Exercise 7.10, using `math.radians` for the angle. Print them for $n = 8$ and confirm the first is directly to the right of the center.

C7.5 Write `ray_hits_circle(origin, direction, center, r)` that reuses C7.2 and keeps only intersections with $t \geq 0$, returning the nearest hit or `None`. Test a ray that points away from the circle and one that starts inside it.

Chapter 8

Right-Triangle Trigonometry

Trigonometry starts as three ratios in a right triangle, and those three ratios are how an angle and a distance become an x and a y. Every sprite that moves at a heading, every camera that tilts, and every shadow that falls uses nothing more than this chapter.

8.1 Core Ideas

The three ratios. For an acute angle θ in a right triangle, name the sides relative to θ: the opposite leg, the adjacent leg, and the hypotenuse (Figure 8.1). The mnemonic SOH CAH TOA records the definitions.

Sine, cosine and tangent

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin\theta}{\cos\theta}$$

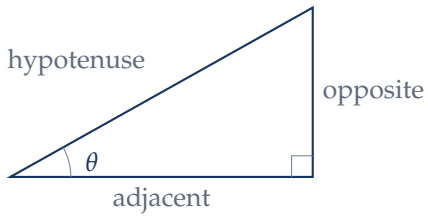


Figure 8.1: The sides of a right triangle named relative to the angle θ.

Why the ratios are well defined. Two right triangles with the same acute angle are similar (AA), so their side ratios agree. The sine of 30° is $\frac{1}{2}$ in every right triangle that contains a 30° angle, whatever its size.

Exact values. The special triangles from Chapter 3 give exact values for 30°, 45° and 60°.

Exact values

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

Inverse functions. arcsin, arccos and arctan (also written $\sin^{-1}$ and so on) recover an angle from a ratio. In code, `math.asin` and friends return radians; multiply by $180/\pi$ or use `math.degrees` until Chapter 9 makes radians native.

Solving a right triangle. Given any two of the five quantities (two legs, hypotenuse, two acute angles) other than the two angles alone, the rest follow from the ratios, the Pythagorean theorem and the fact that the acute angles sum to 90°. Angle of elevation is measured up from the horizontal, angle of depression down.

Pythagorean identity. Dividing $a^2 + b^2 = c^2$ by c^2 gives $\sin^2\theta + \cos^2\theta = 1$ for every angle, the first trigonometric identity and the reason a direction vector built from $(\cos\theta, \sin\theta)$ always has length 1.

Headings in code. A move of distance d at angle θ from the positive x-axis changes position by $(d\cos\theta, d\sin\theta)$. The reverse, angle from (dx, dy), needs `math.atan2(dy, dx)`, not `atan(dy/dx)`: the two-argument version knows the quadrant and never divides by zero.

8.2 Worked Examples

Example 8.1 A right triangle has opposite side 5 and adjacent side 12 relative to θ. Find the three ratios.

Solution. The hypotenuse is $\sqrt{25 + 144} = 13$. So $\sin\theta = \frac{5}{13}$, $\cos\theta = \frac{12}{13}$ and $\tan\theta = \frac{5}{12}$.

Example 8.2 The hypotenuse is 20 and one acute angle is 35°. Find the side opposite that angle.

Solution. $\sin 35^\circ = x/20$, so $x = 20\sin 35^\circ \approx 20(0.5736) = 11.47$.

Example 8.3 The legs are 7 (opposite) and 4 (adjacent). Find the angle.

Solution. $\tan\theta = 7/4 = 1.75$, so $\theta = \arctan 1.75 \approx 60.26^\circ$.

8.3 Exercises

8.1 A right triangle has legs 8 and 15. Find the sine, cosine and tangent of the angle opposite the side of length 8.

8.2 Give exact values: $\sin 30^\circ$, $\cos 45^\circ$, $\tan 60^\circ$.

8.3 The hypotenuse is 10 and one angle is 25°. Find the opposite and adjacent sides to two decimals.

8.4 The adjacent side is 9 and the angle is 50°. Find the opposite side and the hypotenuse.

8.5 The opposite side is 6 and the hypotenuse is 10. Find the angle to two decimals.

8.6 A 20-foot ladder makes a 65° angle with the ground. How high up the wall does it reach, and how far is its base from the wall?

8.7 From 150 m away, the angle of elevation to the top of a building is 32°. How tall is the building?

8.8 A sprite moves 12 units at 30° above the positive x-axis. Find the change in x and in y, exactly and to two decimals.

8.9 The vector from $(0, 0)$ to $(-4, 3)$ has $\arctan(3/(-4)) \approx -36.87^\circ$, but that is the wrong direction. Explain why and give the correct angle.

8.10 Verify $\sin^2\theta + \cos^2\theta = 1$ for the angle opposite the side of length 8 in an 8-15-17 triangle.

8.11 Solve the right triangle with hypotenuse 13 and one leg 5: find the other leg and both acute angles.

8.12 **(proof)** Explain why $\sin\theta$ depends only on θ and not on the size of the right triangle that contains it.

8.4 Programming Exercises

C8.1 Write `sin_deg(t)`, `cos_deg(t)` and `tan_deg(t)` that take degrees, and test them against the exact values in Exercise 8.2 with `math.isclose`.

C8.2 Write `solve_right_triangle(**known)` that accepts any two of `a`, `b`, `c`, `A`, `B` (legs, hypotenuse, angles opposite them in degrees), refuses the two angles alone, and returns all five. Test with Exercises 8.3, 8.4 and 8.11.

C8.3 Write `polar_to_cartesian(d, deg)` and `cartesian_to_polar(dx, dy)` using `math.atan2`. Assert they round-trip for points in all four quadrants, and reproduce Exercise 8.9 to show where `math.atan` goes wrong.

C8.4 Write `height_from_elevation(distance, deg)` for an angle of elevation, and then the function `distance_from_depression(height, deg)` for an angle of depression. Verify Exercise 8.7.

C8.5 Build the table of sin and cos for 0°, 30°, 45°, 60° and 90° using only `math.sqrt` and the special-triangle ratios, no `math.sin`, then compare each entry with `math.sin` and `math.cos`.

Chapter 9

The Unit Circle and Radians

Degrees are a human convention; radians are the angle measure the mathematics prefers and every graphics, audio and physics API expects. The unit circle extends sine and cosine from acute angles to every angle, and turns an angle into a direction vector in one step.

9.1 Core Ideas

Radians. One radian is the angle whose arc on a circle equals the radius. A full turn is 2π radians, so $180^\circ = \pi$ radians.

Degrees, radians and the unit circle

$$\text{radians} = \text{degrees} \cdot \frac{\pi}{180}, \qquad \text{degrees} = \text{radians} \cdot \frac{180}{\pi}, \qquad P(\theta) = (\cos\theta,\ \sin\theta)$$

The unit circle. On the circle of radius 1 centered at the origin, the point at angle θ measured counterclockwise from the positive x-axis is $(\cos\theta, \sin\theta)$ (Figure 9.1). This defines sine and cosine for every angle, not only acute ones, and $\tan\theta$ is $\sin\theta / \cos\theta$ wherever $\cos\theta \neq 0$.

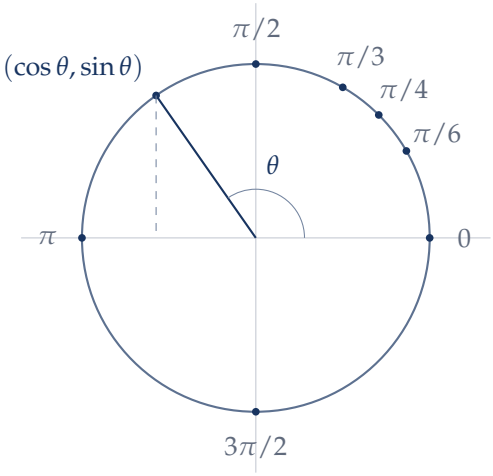


Figure 9.1: The unit circle. Each angle θ gives the point $(\cos\theta, \sin\theta)$; the marked angles are the ones to know by heart.

Signs by quadrant. Cosine is the x-coordinate and sine the y-coordinate, so in quadrant I both are positive, in II only sine, in III neither, in IV only cosine. Tangent is positive in I and III.

Reference angles. The reference angle is the acute angle between the terminal side and the x-axis. The sine and cosine of any angle equal those of its reference angle up to sign, so the table of exact values from Chapter 8 covers the whole circle.

Periodicity and negative angles. Adding 2π (a full turn) changes nothing: $\sin(\theta + 2\pi) = \sin\theta$. Negative angles go clockwise, and $\cos(-\theta) = \cos\theta$ while $\sin(-\theta) = -\sin\theta$.

Arc length and sector area in radians. With θ in radians the degree formulas lose their fraction: $s = r\theta$ and $A = \tfrac{1}{2}r^2\theta$. This simplification is the reason mathematics uses radians.

Angles in code. `math.sin`, `Math.sin`, canvas `arc` and every physics engine take radians. Two further operations come up constantly: wrapping an angle into $[0, 2\pi)$ or $[-\pi, \pi)$, and finding the shortest signed turn from one heading to another, which is never more than half a turn.

9.2 Worked Examples

> **Example 9.1** Convert 150° to radians and $\pi/4$ to degrees.
> *Solution.* $150 \cdot \pi/180 = 5\pi/6$. $\pi/4 \cdot 180/\pi = 45^\circ$.

> **Example 9.2** Find $\sin(5\pi/6)$ and $\cos(5\pi/6)$ exactly.
> *Solution.* $5\pi/6$ is in quadrant II with reference angle $\pi/6$. So $\sin(5\pi/6) = \sin(\pi/6) = \tfrac{1}{2}$, and $\cos(5\pi/6) = -\cos(\pi/6) = -\tfrac{\sqrt{3}}{2}$.

> **Example 9.3** A circle has radius 4. Find the arc length for a central angle of 2.5 radians.
> *Solution.* $s = r\theta = 4(2.5) = 10$.

9.3 Exercises

9.1 Convert to radians, exactly: 30°, 45°, 90°, 120°, 270°, 360°.

9.2 Convert to degrees: $\pi/3$, $3\pi/4$, $7\pi/6$, $5\pi/3$, and 1 radian (to two decimals).

9.3 Give the unit circle point at $\pi/3$, at $3\pi/4$, and at π.

9.4 Evaluate exactly: $\sin(7\pi/6)$, $\cos(5\pi/3)$, $\tan(3\pi/4)$, $\sin(3\pi/2)$.

9.5 Find the reference angle and quadrant of $4\pi/3$ and of $-\pi/4$.

9.6 Find every θ in $[0, 2\pi)$ with $\cos\theta = -\tfrac{\sqrt{2}}{2}$.

9.7 A circle has radius 6. Find the arc length and sector area for a central angle of $\pi/3$.

9.8 A wheel of radius 0.5 m turns at 3 radians per second. How fast does a point on its rim move?

9.9 Normalize $7\pi/2$ to $[0, 2\pi)$, and $-\pi/3$ as well.

9.10 What is the shortest signed rotation from a heading of 350° to 10°? Give it in degrees and radians, and say why -340° is the wrong answer.

9.11 A point orbits on a circle of radius 100 px centered at screen $(400, 300)$, with y pointing down. At $\theta = \pi/6$ measured counterclockwise in math coordinates, what are its screen coordinates?

9.12 **(proof)** Show that the degree formula $s = \frac{\theta}{360} \cdot 2\pi r$ becomes $s = r\theta$ when θ is converted to radians.

9.4 Programming Exercises

C9.1 Write `deg_to_rad(d)` and `rad_to_deg(r)` using only `math.pi`, and assert they agree with `math.radians` and `math.degrees` for every integer degree from -720 to 720.

C9.2 Write `unit_circle_point(theta)`. Assert $x^2 + y^2 = 1$ within tolerance for 1000 random angles, and check Exercise 9.3 against exact values.

C9.3 Write `wrap_2pi(theta)` into $[0, 2\pi)$ and `wrap_pi(theta)` into $[-\pi, \pi)$. Test with Exercise 9.9 and with angles far outside one turn, such as $100\pi + 1$.

C9.4 Write `shortest_turn(from_deg, to_deg)` returning the signed turn in $(-180, 180]$. Test with Exercise 9.10, with a turn of exactly 180°, and with two equal headings.

C9.5 Write `orbit_position(t, omega, r, center)` for a point orbiting at angular speed `omega` radians per second, returning screen coordinates with y down. Confirm Exercise 9.11 at the right `t`, and print one full orbit at 12 steps to see the direction of travel.

Chapter 10

Vectors in Two Dimensions

A vector is a displacement, a velocity, a force, or a direction, all stored as the same pair of numbers. Adding them moves things, normalizing them keeps speeds honest, and the dot product answers the questions a game asks most: how far apart, how aligned, and is it in front of me.

10.1 Core Ideas

Vectors and components. A vector $\mathbf{v} = \langle x, y \rangle$ has magnitude and direction but no position. The vector from point A to point B is $B - A$, component by component. A point's position vector is the vector from the origin to it.

Addition and scaling. $\langle a, b \rangle + \langle c, d \rangle = \langle a + c, b + d \rangle$, pictured tip to tail. $k\langle a, b \rangle = \langle ka, kb \rangle$ scales the length by $|k|$ and reverses direction when $k < 0$. Subtraction $\mathbf{u} - \mathbf{v}$ is the vector from the tip of $\mathbf{v}$ to the tip of $\mathbf{u}$. Position plus velocity times time is this arithmetic.

Magnitude and unit vectors. $|\langle x, y \rangle| = \sqrt{x^2 + y^2}$ by Chapter 3. The unit vector in the direction of $\mathbf{v}$ is $\mathbf{v}/|\mathbf{v}|$, which has length 1; multiplying it by a speed gives a velocity with that exact speed. The zero vector has no direction and cannot be normalized. The direction angle is $\operatorname{atan2}(y, x)$.

The dot product. Defined by components and equal to a product of lengths and the cosine of the angle between the vectors. Its sign says whether the angle is acute (positive), right (zero) or obtuse (negative), with no trigonometry needed.

Dot product

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y = |\mathbf{a}|\,|\mathbf{b}| \cos\theta, \qquad \cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|\,|\mathbf{b}|}$$

Projection. The projection of $\mathbf{a}$ onto $\mathbf{b}$ is the part of $\mathbf{a}$ that points along $\mathbf{b}$: $\operatorname{proj}_{\mathbf{b}} \mathbf{a} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2}\,\mathbf{b}$ (Figure 10.1). Its length, the scalar projection, is $\mathbf{a} \cdot \mathbf{b}/|\mathbf{b}|$. Sliding along a wall keeps only the projection of the velocity onto the wall.

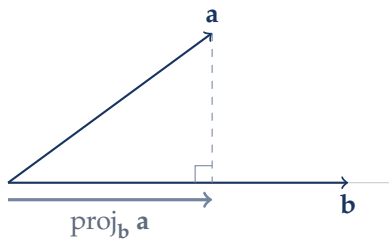


Figure 10.1: The projection of **a** onto **b** is the shadow **a** casts on the line of **b**.

Perpendicular vectors. In two dimensions, $\langle -y, x\rangle$ is perpendicular to $\langle x, y\rangle$ and has the same length; it is the counterclockwise 90° rotation from Chapter 5. The dot product of perpendicular vectors is zero.

Reflection. A velocity $\mathbf{v}$ bouncing off a surface with unit normal $\mathbf{n}$ becomes $\mathbf{v} - 2(\mathbf{v}\cdot\mathbf{n})\mathbf{n}$: the component along the normal is reversed and the rest is kept.

10.2 Worked Examples

Example 10.1 Let $\mathbf{a} = \langle 3, -1\rangle$ and $\mathbf{b} = \langle -2, 4\rangle$. Find $\mathbf{a}+\mathbf{b}$, $2\mathbf{a}-\mathbf{b}$ and $|\mathbf{a}|$.

Solution. $\mathbf{a}+\mathbf{b} = \langle 1, 3\rangle$. $2\mathbf{a}-\mathbf{b} = \langle 6, -2\rangle - \langle -2, 4\rangle = \langle 8, -6\rangle$. $|\mathbf{a}| = \sqrt{9+1} = \sqrt{10}$.

Example 10.2 Find the angle between $\langle 1, 2\rangle$ and $\langle 3, 1\rangle$.

Solution. The dot product is $3+2=5$. The lengths are $\sqrt{5}$ and $\sqrt{10}$, so $\cos\theta = 5/\sqrt{50} = 1/\sqrt{2}$ and $\theta = 45^\circ$.

Example 10.3 Project $\langle 3, 4\rangle$ onto $\langle 1, 1\rangle$.

Solution. The dot product is 7 and $|\langle 1,1\rangle|^2 = 2$, so the projection is $\frac{7}{2}\langle 1, 1\rangle = \langle 3.5, 3.5\rangle$.

10.3 Exercises

10.1 For $\mathbf{u} = \langle 2, -3\rangle$ and $\mathbf{v} = \langle -1, 5\rangle$, find $\mathbf{u}+\mathbf{v}$, $\mathbf{u}-\mathbf{v}$, $3\mathbf{u}$ and $|\mathbf{u}|$.

10.2 Find the displacement vector from $P(1, 4)$ to $Q(-2, 8)$, its magnitude, and the unit vector in its direction.

10.3 Find the unit vector in the direction of $\langle 6, 8\rangle$, and the vector of length 15 in that direction.

10.4 Compute $\langle 3, 4\rangle \cdot \langle 4, -3\rangle$, $\langle 1, 2\rangle \cdot \langle 2, 1\rangle$ and $\langle -1, 2\rangle \cdot \langle 3, 1\rangle$, and state what each sign says about the angle.

10.5 Find the angle between $\langle 1, 0\rangle$ and $\langle 1, \sqrt{3}\rangle$.

10.6 Find the angle between $\langle 2, 2\rangle$ and $\langle -1, 1\rangle$.

10.7 Project $\langle 5, 3\rangle$ onto $\langle 2, 0\rangle$ and give the scalar projection. Then project $\langle 2, 6\rangle$ onto $\langle 1, 1\rangle$.

10.8 Give a vector perpendicular to $\langle 3, 7\rangle$ and verify it with the dot product.

10.9 Find the direction angle of $\langle -4, -4 \rangle$.

10.10 A character at $(10, 2)$ faces direction $\langle 1, 0 \rangle$ and an enemy stands at $(14, 5)$. Find the vector to the enemy, its dot product with the facing direction, and the angle between them. Is the enemy inside a view cone of half-angle 45°?

10.11 A ball with velocity $\langle 3, -4 \rangle$ hits a horizontal floor with unit normal $\langle 0, 1 \rangle$. Find its velocity after the bounce.

10.12 **(proof)** Using components, prove that $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$, and that $\langle -y, x \rangle$ is perpendicular to $\langle x, y \rangle$ with the same length.

10.4 Programming Exercises

C10.1 Write `add`, `sub`, `scale`, `magnitude`, `normalize` (raising on the zero vector) and `dot` on coordinate tuples. Test with Exercises 10.1 to 10.4.

C10.2 Write `angle_between(a, b)` in degrees using `math.acos`, clamping the cosine to $[-1, 1]$ first. Test with Exercises 10.5 and 10.6, then explain in a comment why the clamp is needed when `a` and `b` are parallel.

C10.3 Write `project(a, b)` and `perp(v)`. Test with Exercises 10.7 and 10.8, and assert that `dot(perp(v), v)` is zero for 100 random vectors.

C10.4 Write `in_view_cone(pos, facing, target, half_angle_deg)`. Test with Exercise 10.10, then move the target around a circle of radius 5 about the character and print which angles are visible.

C10.5 Write `reflect(v, n)` and simulate a ball inside a box for 500 steps, bouncing off all four walls. Assert that the speed never changes, and explain which of the chapter's facts guarantees that.

Chapter 11

Course Project

The project is a 2D scene, on an HTML canvas, in SVG, or in matplotlib, where you translate, rotate and scale shapes by typing values, click to find out what you hit, intersect a line with a circle, and read the angle between two vectors. If the shapes behave wrong, the math is wrong, which is the point.

Build it in six stages, each with its own acceptance tests. Do not start a stage until the previous one passes. All geometry lives in math coordinates and is converted to screen coordinates only when drawing, using `to_screen` from Chapter 4.

11.1 The Six Stages

Stage	Build	Reuses	Done when
1. Scene and axes	An 800 by 600 drawing surface with the math origin at the center and y up; draw the axes and a labeled grid	C4.4	The math point $(0, 0)$ draws at the center and $(100, 100)$ draws up and to the right
2. Shapes and transforms	Shapes as vertex lists; translate, rotate (any angle) and scale from typed values, each as a 3×3 matrix, applied in the order typed	C5.1, C5.4, C9.2	Rotating a square by 90° four times returns its vertices within 10^{-9}, and swapping the order of a rotate and a translate visibly moves the result
3. Hit testing	Click a point: report which shape contains it, using the squared-distance test for circles and a same-side test for convex polygons (the point is inside when it is on the same side of every edge)	C4.3, C6.1	For 1000 random points and a convex polygon, the same-side test agrees with a ray-crossing test you write independently

Table 11.1: Stages 1 to 3 of the geometry scene, what each reuses, and its acceptance test.

Stage	Build	Reuses	Done when
4. Circle and line	Draw one circle and one segment; mark the intersection points and report their count	C7.2, C7.5	Moving the segment reports 0, 1 or 2 points correctly, and the tangent case shows exactly one mark
5. Vector readout	Two vectors from the origin, set by typing or dragging; display magnitudes, dot product, angle and the projection of one onto the other	C10.1 to C10.3	Perpendicular vectors show 90.00°, parallel vectors show 0.00° without a `NaN`, and the drawn projection lands on the line of the second vector
6. Motion	Animate a shape turning from one heading to another by the shortest turn, and a point orbiting a circle, all in radians	C9.3, C9.4, C9.5	Turning from 350° to 10° passes through 0°, never through 180°, and the orbit runs counterclockwise on screen when the math angle increases

Table 11.2: Stages 4 to 6 of the geometry scene.

11.2 Acceptance Checklist

- ❏ Every transform is a matrix, and the scene composes them by multiplication rather than by special-casing each kind.
- ❏ Every numeric exercise answer in Chapters 4, 5, 6, 7 and 10 that involves points or vectors can be reproduced by typing the inputs into the scene.
- ❏ A test file runs the agreement checks from Stages 2, 3 and 5 with one command and prints a pass count.
- ❏ No function in the project calls `math.atan`; every angle recovered from coordinates uses `math.atan2`.
- ❏ A short README explains how to run it and which chapter each module came from.

11.3 Stretch Goals

Add a concave polygon and make the ray-crossing test the one the scene uses for it. Add a bounding-box broad phase so that hit testing skips shapes whose boxes miss the click, and show the boxes. Add a sector-shaped view cone to a shape and highlight any other shape inside it, using Exercise C10.4.

Exit Review

Solve these twelve problems by hand in one sitting, with no notes, then verify every numeric answer with your scene from Chapter 11 or the chapter functions it is built from. Eleven correct passes the course; fewer means rereading the chapters you missed and retaking it a week later. These answers are deliberately not in the key.

E.1 Write the contrapositive of "If two lines are parallel, then corresponding angles are congruent," and state whether it is true.

E.2 Two parallel lines are cut by a transversal and one angle measures 52°. Find its same-side interior partner and its alternate exterior partner.

E.3 Is a triangle with sides 10, 24, 26 a right triangle? Find its area.

E.4 Find the distance and midpoint between $(-2,-3)$ and $(4,5)$.

E.5 Rotate $(5,-2)$ by 90° counterclockwise about the origin, then reflect the result across the y-axis.

E.6 Find each interior angle and each exterior angle of a regular decagon.

E.7 Find the area of the polygon $(0,0)$, $(6,0)$, $(6,4)$, $(2,7)$ by the shoelace formula.

E.8 A circle has radius 9. Find the arc length and sector area for a central angle of 80°, exactly and to two decimals.

E.9 A 30-foot ladder leans against a wall at 72° to the ground. How high does it reach, to two decimals?

E.10 Convert 210° to radians and give sin and cos of that angle exactly.

E.11 Find the angle between $\langle 3,4\rangle$ and $\langle -8,6\rangle$.

E.12 Find where the line $y = x$ meets the circle $x^2 + y^2 = 50$.

The curriculum's exit check for Course 2 also applies, and the review above covers it:

- ❑ Prove the Pythagorean theorem and the triangle angle sum.
- ❑ Convert the standard angles between degrees and radians without a calculator.
- ❑ Compute distance, midpoint and the dot product of two vectors, and explain what the sign of the dot product tells you.
- ❑ Write a function that rotates a point about the origin by any angle.

Answers to Exercises

Check only after finishing a chapter's full set. A wrong answer goes in your mistake log with the rule you broke. Proof exercises are not answered here; a proof is checked by reading it aloud to someone who has not seen it.

Chapter 1

1.1 Hypothesis: two angles are complementary. Conclusion: their measures add to 90°.

1.2 Converse: if two angles are congruent, they are vertical (false). Inverse: if two angles are not vertical, they are not congruent (false). Contrapositive: if two angles are not congruent, they are not vertical (true). The original is true.

1.3 6 (or 2, 10, any number divisible by 2 but not 4)

1.4 $AC = AB + BC$ and $BD = BC + CD$ by segment addition; $AB = CD$ is given; substitute to get $AC = BD$.

1.5 $\angle 1 + \angle 2 = 180^\circ$ and $\angle 2 + \angle 3 = 180^\circ$; so $\angle 1 + \angle 2 = \angle 2 + \angle 3$; subtract $\angle 2$.

1.6 An angle is a right angle if and only if it measures 90°. If an angle is right then it measures 90°; if an angle measures 90° then it is right.

1.7 True. Converse false: $x = 4$.

1.8 If a triangle is not isosceles, then it does not have two congruent angles. A statement and its contrapositive are equivalent.

1.9 If $n \geq 0$ then the call returns $n!$. Contrapositive: if the call did not return $n!$, then $n < 0$. A wrong value means the precondition was violated (assuming the function is correct).

1.10 $ABCD$ is a rectangle. No; that would be affirming the consequent (assuming the converse).

Chapter 2

2.1 53°; 143°

2.2 $x = 20$; 65°

2.3 $x = 35$; 117° and 63°

2.4 $x = 20$; 70°

2.5 75° and 105°

2.6 60° and 70°

2.7 75°

2.9 The second; alternate interior angles must be congruent, and $62 \neq 118$.

2.10 15°; 270°

Chapter 3

3.1 (a) equilateral, acute; (b) isosceles, right; (c) scalene, acute

3.2 (a) no; (b) yes; (c) no (degenerate)

3.3 $3 < x < 17$

3.4 17

3.5 24

3.6 Yes, $81 + 1600 = 1681 = 41^2$. No, $49 + 81 = 130 < 144$, so obtuse.

3.7 $6\sqrt{2}$; $5\sqrt{3}$ and 10

3.8 70° each

3.9 $EF = 12$; perimeters $2 : 3$

3.10 (a) SSS; (b) ASA; (c) none (SSA)

3.11 5 feet

3.13 About 2202.9 pixels

Chapter 4

4.1 10

4.2 $(-1, 2)$

4.3 $(7, 2)$

4.4 24

4.5 $(x+3)^2 + (y-2)^2 = 16$

4.6 Center $(3, -2)$, radius 5

4.7 On the circle ($4^2 + 3^2 = 25$)

4.8 Slopes: AB and DC both 0, AD and BC both 3. Diagonals AC and BD share the midpoint $(2.5, 1.5)$.

4.9 $(300, 250)$; $(250, -150)$

4.11 Inside; on the boundary; outside

Chapter 5

5.1 $(2, 5)$

5.2 $(4, 1)$; $(-4, -1)$; $(-1, 4)$

5.3 $(-5, -2)$; $(2, -5)$; $(5, 2)$

5.4 $(2, -1)$; $k = 2.5$

5.5 Translation 4 left and 1 up; $(-4, 1)$, $(-2, 1)$, $(-2, 3)$, $(-4, 3)$

5.6 $(0, 4)$

5.7 $(1, 3)$; reversed, $(-1, -3)$; not the same

5.8 Translation, rotation and reflection are isometries; only reflection reverses orientation.

5.9 Order 6, 60°; two lines

5.10 $(50, 0)$; $(0, 50)$. The rightmost transform acts first.

5.11 $(x, y) \mapsto (x, y+6)$, a translation by 6 upward; in general a translation by $2d$ perpendicular to the lines.

Chapter 6

6.1 1080°; 135°; 45°

6.2 12

6.3 36; 35

6.4 50

6.5 120; side 13

6.6 $54\sqrt{3} \approx 93.5$

6.7 14.5

6.8 3 and 9

6.9 15 both ways

6.10 Box from $(-2, -1)$ to $(5, 6)$, area 49; triangle area 19.5; about 0.40 of the box

Chapter 7

7.1 $14\pi \approx 43.98$; $49\pi \approx 153.94$

7.2 $2\pi \approx 6.28$; $6\pi \approx 18.85$

7.3 90°

7.4 70°; 140°

7.5 12; the angle opposite the diameter is a right angle, so Pythagoras applies.

7.6 10

7.7 6

7.8 Miss; tangent at $(0, 5)$; secant at $(4, 3)$ and $(-4, 3)$

7.9 $(3, 4)$ and $(4, 3)$

7.10 45°; $30\pi \approx 94.25$ pixels

7.11 $350\pi \approx 1099.6$ m

Chapter 8

8.1 $\frac{8}{17}$, $\frac{15}{17}$, $\frac{8}{15}$

8.2 $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$, $\sqrt{3}$

8.3 4.23; 9.06

8.4 10.73; 14.00

8.5 36.87°

8.6 18.13 ft; 8.45 ft

8.7 About 93.7 m

8.8 $(6\sqrt{3}, 6) \approx (10.39, 6)$

8.9 arctan cannot tell $(-4, 3)$ from $(4, -3)$; add 180° for quadrant II: 143.13°.

8.10 $\frac{64}{289} + \frac{225}{289} = 1$

8.11 Other leg 12; angles 22.62° and 67.38°

Chapter 9

9.1 $\pi/6$, $\pi/4$, $\pi/2$, $2\pi/3$, $3\pi/2$, 2π

9.2 60°, 135°, 210°, 300°, 57.30°

9.3 $(\frac{1}{2}, \frac{\sqrt{3}}{2})$; $(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$; $(-1, 0)$

9.4 $-\frac{1}{2}$; $\frac{1}{2}$; -1; -1

9.5 $\pi/3$ in quadrant III; $\pi/4$ in quadrant IV

9.6 $3\pi/4$ and $5\pi/4$

9.7 $2\pi \approx 6.28$; $6\pi \approx 18.85$

9.8 1.5 m/s

9.9 $3\pi/2$; $5\pi/3$
9.10 $+20^\circ = \pi/9$; -340° is the long way around
9.11 $(486.6, 250)$

Chapter 10

10.1 $\langle 1, 2\rangle$; $\langle 3, -8\rangle$; $\langle 6, -9\rangle$; $\sqrt{13}$
10.2 $\langle -3, 4\rangle$; 5; $\langle -\frac{3}{5}, \frac{4}{5}\rangle$
10.3 $\langle 0.6, 0.8\rangle$; $\langle 9, 12\rangle$
10.4 0 (right angle); 4 (acute); -1 (obtuse)
10.5 60°
10.6 90°
10.7 $\langle 5, 0\rangle$, scalar projection 5; $\langle 4, 4\rangle$
10.8 $\langle -7, 3\rangle$ (or $\langle 7, -3\rangle$); $-21 + 21 = 0$
10.9 225° (equivalently -135°)
10.10 $\langle 4, 3\rangle$; dot product 4; $\cos\theta = 0.8$, $\theta \approx 36.87^\circ$; yes
10.11 $\langle 3, 4\rangle$